```
clc;
clear;
close all;
[file,path] = uigetfile('*.wav','Select Speech Signal');
if isequal(file,0)
error('No speech file selected');
end

[x,fs_in] = audioread(fullfile(path,file));
x = mean(x,2); % Stereo to mono

fs = 16000; % Hearing-aid sampling rate
if fs_in ~= fs
x = resample(x,fs,fs_in);
end

figure;
subplot(3,2,1);
plot(x)
grid on
xlabel('Samples')
ylabel('Amplitude')
title('(a) Input Speech Signal')
[file_csv,path_csv] = uigetfile('*.csv','Select OHHR Audiogram CSV');
if isequal(file_csv,0)
error('No audiogram file selected');
end

ohhr = readtable(fullfile(path_csv,file_csv));
subjectID = 1;
earType = "Right";

idx = ohhr.Subject_ID == subjectID & ohhr.Ear == earType;
```

```
freq_aud = ohhr.Frequency_Hz(idx)';
audiogram = ohhr.Threshold_dBHL(idx)';
gain_dB = audiogram;
gain_lin = 10.^(gain_dB/20);

subplot(3,2,2);
plot(freq_aud,audiogram,'-o','LineWidth',2)
set(gca,'XScale','log','YDir','reverse')
grid on
xlabel('Frequency (Hz)')
ylabel('Hearing Level (dB HL)')
title('(b) OHHR Audiogram')
N = 128; % FIR filter order

bands = [ ...
0 500;
500 1000;
1000 2000;
2000 4000;
4000 8000];

numBands = size(bands,1);
h = cell(numBands,1);

for k = 1:numBands
f1 = bands(k,1);
f2 = bands(k,2);

if f1 == 0
Wn = f2/(fs/2);
h{k} = fir1(N, Wn, 'low');
elseif f2 == fs/2
```

```
Wn = f1/(fs/2);
h{k} = fir1(N, Wn, 'high');
else
Wn = [f1 f2]/(fs/2);
h{k} = fir1(N, Wn, 'bandpass');
end
end

subplot(3,2,3); hold on;
for k = 1:numBands
[H,f] = freqz(h{k},1,2048,fs);
plot(f,20*log10(abs(H)))
end
grid on
xlim([0 fs/2])
ylim([-80 5])
xlabel('Frequency (Hz)')
ylabel('Magnitude (dB)')
title('(c) Filter Bank Responses')
band_gain = zeros(numBands,1);

for k = 1:numBands
f1 = bands(k,1);
f2 = bands(k,2);

idx_band = freq_aud >= max(f1,1) & freq_aud <= f2;
band_gain(k) = mean(gain_lin(idx_band));
end

subplot(3,2,4);
stem(1:numBands,20*log10(band_gain),'LineWidth',1.5)
grid on
xlabel('Band Index')
```

```
ylabel('Gain (dB)')
title('(d) Subband Gains from Audiogram')
x_band = zeros(length(x),numBands);

for k = 1:numBands
x_band(:,k) = filter(h{k},1,x) * band_gain(k);
end

y = sum(x_band,2);
y = y / max(abs(y)); % Normalize

subplot(3,2,5);
plot(y)
grid on
xlabel('Samples')
ylabel('Amplitude')
title('(e) Hearing Aid Output Signal')
audiowrite('hearing_aid_output_OHHR.wav', y, fs);
disp('✔ Hearing-aid output saved as hearing_aid_output_OHHR.wav');
```